

Question	Marking details			Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	Other nucleic acids involved in protein synthesis:			6		3	9		
	Nucleic acid	role	Where it functions						
	DNA	provides the {genetic code/ template} for production of {all RNAs/ mRNA/ provides the genetic code for sequence of amino acids}	nucleus						
	rRNA	To form ribosomes/ the site of translation	cytoplasm/ rough ER						
	mRNA	Produced in transcription and used in translation/ transfers genetic code (from nucleus to ribosomes) / owtte	cytoplasm/ ribosomes						
	tRNA	delivery of amino acids to the site of protein synthesis	cytoplasm						
	Section C: How tRNA activation works: • ATP provides energy for the reaction • Two phosphates are released • Amino acid reacts with ATP to form activated amino acid. Amino acid is transferred to tRNA forming activated {tRNA/ amino acid}								

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		AO1	AO2	AO3	Total	Maths	Prac
	7-9 marks Indicative content of this level is... Detailed description of role of four types of nucleic acid Identification of location of action of all nucleic acids Detailed description of tRNA activation <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is... Two from: description of role of some types of nucleic acid Identification of location of action of most nucleic acids description of tRNA activation <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is... Any correct statements from description of role of some types of nucleic acid Identification of location of action of most nucleic acids description of tRNA activation <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
	Question 7 total	6	0	3	9	0	0